YELLOW REPORT

In primary mediastinal nonseminomatous germ cell tumors, **BEP and VIP have similar survival and cure rates** [1], but **VIP avoids bleomycin and is preferred in patients with pulmonary risk** [2]. VIP causes more hematologic toxicity and requires G-CSF support [1], but **does not significantly increase non-hematologic morbidity** compared with BEP [1]. For a 20-year-old with asthma, **VIP is reasonable** to mitigate bleomycin-related lung injury risk, though close pulmonary monitoring is still essential [2]. In this cohort, any regimen — BEP or VIP — demands pulmonary monitoring because the mediastinal primary confers a high risk of post-chemotherapy complications; pre- and post-treatment spirometry with DLCO and serial CXR/CT are prudent to detect and manage bleomycin pneumonitis or postoperative complications [3] [4].

Comparative efficacy: overall survival and cure rates

- **Overall survival**: BEP and VIP have similar overall survival in disseminated germ cell tumors, with no significant difference in long-term outcomes [1] [5].
- **Cure rates**: Both regimens achieve comparable cure rates, particularly in poor-risk disease, with no statistically significant difference in complete response or progression-free survival [1] [5].
- **Mediastinal primary**: Primary mediastinal nonseminomatous germ cell tumors have worse outcomes than gonadal primaries, but this is independent of the regimen used; both BEP and VIP are standard options [2] [3].

Pulmonary complications and treatment-related morbidity

- **Pulmonary toxicity**: BEP causes significant pulmonary toxicity, including pneumonitis and fibrosis, which is particularly concerning in mediastinal tumors and patients with pre-existing lung disease [6] [7].
- **Postoperative complications**: In primary mediastinal nonseminomatous germ cell tumors, patients receiving BEP have higher rates of postoperative respiratory failure, pneumonia, and mortality compared with VIP, likely due to bleomycin-induced lung injury [6].
- **VIP morbidity**: VIP has higher hematologic toxicity (neutropenia, thrombocytopenia) and requires G-CSF support, but **does not increase non-hematologic morbidity** compared with BEP [1] [2].

Clinical guidelines and recommendations

NCCN and other guidelines **recommend BEP as first-line therapy** for primary mediastinal nonseminomatous germ cell tumors [2]. VIP is **reserved for patients with contraindications to bleomycin** (e.g. pulmonary disease, age > 50, or high bleomycin toxicity risk) [2]. For a 20-year-old with asthma, **VIP is a reasonable alternative** to avoid bleomycin-related pulmonary toxicity, though close monitoring is still required [2] [6].

Summary of evidence

Outcome	BEP	VIP
Overall survival	Similar to VIP [1]	Similar to BEP [5]
Cure rates	Comparable to VIP [1]	Comparable to BEP [5]
Pulmonary complications	Significant; especially in mediastinal tumors [6] [7]	Avoided; no bleomycin [2]
Treatment-related morbidity	Hematologic and non-hematologic toxicity	Primarily hematologic toxicity [1]
Clinical guidelines	First-line therapy [2]	Alternative for bleomycin contraindications [2]

Sources:

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2. NCCN guidelines® insights: testicular cancer, version 2.2025. Journal of the National Comprehensive Cancer Network, 2025
3. Mediastinal germ cell tumors: updates in diagnosis and management. Surgical Oncology Clinics of North America, 2020
4. Primary mediastinal germ cell tumors: a real-world analysis of clinical characteristics, treatment, and survival outcomes from two tertiary cancer centers in India. JCO Global Oncology, 2025
5. A randomized phase III study comparing standard dose BEP with sequential high-dose cisplatin, etoposide, and ifosfamide (VIP) plus stem-cell support in males with poor-prognosis germ-cell cancer. An intergroup study of EORTC, GTCSG, and grupo germinal (EORTC 30974). Annals of Oncology, 2011
6. Perioperative morbidity and mortality associated with bleomycin in primary mediastinal nonseminomatous germ cell tumor (...). ASCO, 2016
7. Pulmonary function in patients with germ cell cancer treated with bleomycin, etoposide, and cisplatin. Journal of Clinical Oncology, 2016